Naive Bayes Intuition

We represent a text document as if it were a **bag of words**

- an unordered set of words with their position ignored
- keeping only their frequency in the document

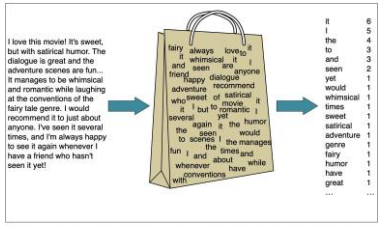

Figure 4.1 Intuition of the multinomial naive Bayes classifier applied to a movie review. The position of the words is ignored (the *bag-of-words* assumption) and we make use of the frequency of each word.

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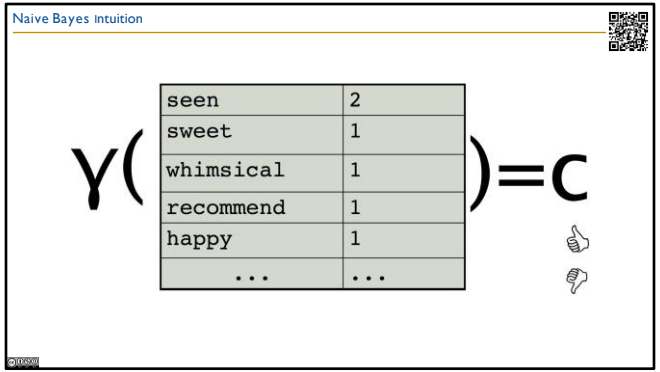
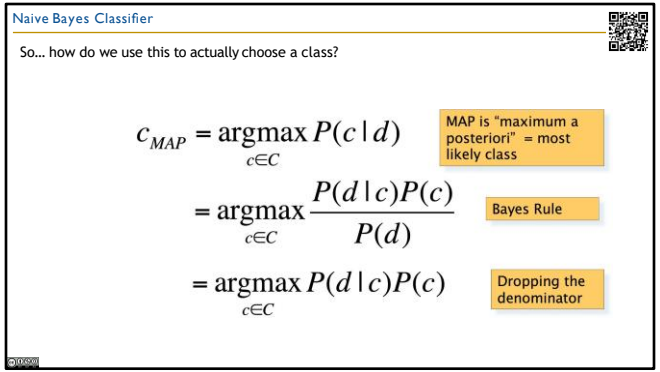


Organize corpus into unique values and their repetition counts

Bayes' Rule Applied to Documents and Classes

For a document *d* and a class *c*

$$P(c|d) = \frac{P(d|c)P(c)}{P(d)}$$



Argmax : in an array input, returns the index position of the value that maximizes the function output (first equation)

- Given document d what is the probability the class is C? (question)
    - Identify which class yields the highest probability based on given information

(third equation)

- Remove denominator
    - Denominator never changes when performing multiple tests
    - Denominator = all available outputs

Naive Bayes Classifier

"Likelihood" "Prior"

$$c_{MAP} = \underset{c \in C}{\operatorname{argmax}} P(d \mid c) P(c)$$

$$= \underset{c \in C}{\operatorname{argmax}} P(x_1, x_2, \ldots, x_n \mid c) P(c)$$

Document d represented as features x1..xn

Second expression is the expanded output of the first one

Naive Bayes Classifier

$$P(x_1, x_2, \ldots, x_n \mid c)$$

To calculate the above likelihood, we need to make some assumptions…

- **Bag of Words assumption:** Assume position doesn't matter
- **Conditional Independence:** Assume the feature probabilities $P(x_i \mid c)$ are independent given the class c

$$P(x_1, x_2, \ldots, x_n \mid c) = P(x_1|c) \cdot P(x_2|c) \cdots P(x_n|c)$$

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Naive Bayes Classifier

$$c_{MAP} = \text{argmax}_{c \in C} P(x_1, x_2, \ldots, x_n \mid c) P(c)$$

$$c_{NB} = \text{argmax}_{c \in C} P(c) \prod_{x \in X} P(x \mid c)$$

Naive Bayes Classifier

To apply the Naive Bayes classifier to text, we need to consider word positions, by simply walking an index through every word position in the document

positions ← all word positions in a test document

$$c_{NB} = \text{argmax}_{c \in C} P(c) \prod_{i \in \text{positions}} P(w_i \mid c)$$

- Position of words would be replaced with words in vocabulary
- Argmax performs multiplication of probability on each class
    - Multiply probability with each class
    - Class yields highest prob will be used

Problem?

What is the problem here?

$$c_{NB} = \text{argmax}_{c \in C} P(c) \prod_{i \in \text{positions}} P(w_i \mid c)$$

Multiplying with lots of probabilities can result in numerical underflow!

So we use logs!

$$c_{NB} = \text{argmax}_{c \in C} \left[\log P(c) + \sum_{i \in \text{positions}} \log P(w_i \mid c) \right]$$

- Numerical underflow : System can’t represent extremely small values
- Using logs can tackle numerical underflow issue

Problem?

Taking log doesn't change the ranking of classes!

- The class with highest probability also has highest log probability!

It's a linear model:

- Just a max of a sum of weights: a linear function of the inputs
- So naive bayes is a linear classifier

$$c_{NB} = \text{argmax}_{c \in C} \left[\log P(c) + \sum_{i \in \text{positions}} \log P(w_i \mid c) \right]$$

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